Digital Forensic Analysis of base-wkstn-01-c-drive.E01

Case ID: CASE-001 **Case Name:** Test1 **Analyst Role:** External DFIR analyst **Report Date (UTC):** 2026-06-04 **Reporting Timezone:** UTC **Evidence Mode:** Strict read-only

Executive Summary

A forensic image of the system volume of host **BASE-WKSTN-01** (Windows 10, domain shieldbase.lan, acquired 2021-09-16 03:10 UTC) was analyzed in read-only mode. The investigation identified the following:

- **Confirmed:** A commercial keylogger family **BlazingTools Perfect Keylogger** is installed on the host at C:\ProgramData\perfmon-k\ with file names that mimic Microsoft Performance Monitor (perfmon-k*). The product is self-identified by the on-disk file install.log. Components and locations are documented in Findings FND-001 to FND-005.
- **Confirmed:** Two of the keylogger launcher binaries (perfmon-k.exe and perfmon-kconfigure.exe) were executed in 2018 (recorded in Amcache) and later deleted by McAfee VirusScan Enterprise on 2019-05-08 and 2019-05-17 (Artemis Trojan detections). Seven supporting keylogger files remain present on disk as of the acquisition date.
- **Probable:** No execution of the residual keylogger files is observed within the available Sysmon telemetry window (2021-07-23 to 2021-09-16); the keylogger appears dormant during that period (FND-005).
- **Confirmed (legitimate):** C:\Windows\subject_srv.exe and C:\Windows\Mnemosyne.sys observed in Prefetch and Sysmon for 2021-09-16 are the F-Response Subject Service and its kernel driver - the DFIR remote-acquisition tool that was used by responders to acquire this very image. These are not malware.
- **Probable false positive:** 84 McAfee GenericRXIT-PW!264609C34194 detections (2019-10-22 to 2019-12-19) against `{GUID}-78.0.3904.70_77.0.3865.120_chrome_updater.exe` in user mhill Local\Temp are most consistent with the documented AV false-positive issue against legitimate Google Chrome 78 auto-updaters.

There is no current evidence of active malware execution, lateral movement, credential theft, or exfiltration in the analyzed artifacts. The principal residual risk on this host is the still-installed Perfect Keylogger component set, which should be removed.

Evidence Examined

Evidence ID	Source Path	Type	Size (Bytes)	MD5	SHA1	Verification
EVID-001	/home/sansforensics/Desktop/cases/Test1/base-wkstn-01-c-drive.E01	E01 volume image (NTFS C: drive only; no MBR/GPT)	33,833,353,216	fd463e4eb6b5744c84b5ad138e7fec7f	c4e37e7fe2d95db50d3c5a8687a5cf6aad101503	ewfverify SUCCESS

Acquisition metadata (from ewfinfo): Acquisition date 2021-09-16 03:10:21; acquired by ADI4.2.0.13 (FTK Imager format), sector size 512, media size 31 GiB.

Mount state: Read-only NTFS via ewfmount + loop (/mnt/wkstn01, options ro,noatime,loop,allow_other per findmnt). See ./analysis/mounts.csv.

Tools and Versions

Tool	Version	Purpose
ewfinfo / ewfverify / ewfmount	libewf 20140816	Image metadata + integrity + read-only mount
The Sleuth Kit (fsstat)	system	NTFS volume metadata
EvtxECmd (Eric Zimmerman)	latest in /opt/zimmermantools	Windows event log parsing
AmcacheParser (Eric Zimmerman)	1.5.2.0	Amcache parsing
AppCompatCacheParser (Eric Zimmerman)	1.5.1.0	ShimCache parsing
pyscca (libscca-python)	20250915	Prefetch parsing
Python 3.12	system	Custom parsing scripts
weasyprint	68.1	Report PDF rendering
PyMuPDF (fitz)	1.27.2	PDF visual sanity check

Host Profile (Observed)

- **Computer name:** base-wkstn-01 / base-wkstn-01.shieldbase.lan
- **Operating system family:** Windows 10 (Sysmon, NTFS volume serial 82424B85424B7CC3)
- **Domain:** shieldbase.lan
- **User profiles present:** Administrator, Administrator.BASE-WKSTN-01, administrator.shieldbase, cbarton-a, mhill, rsydow-a, Public, Default
- **AV / EDR on disk:** McAfee VirusScan Enterprise (+ McAfee Agent), Windows Defender (largely passive), Velociraptor (DFIR endpoint agent, installed by responder)

Timeline of Confirmed Activity

All timestamps in UTC. McAfee OnAccessScanLog timestamps are recorded in host local time without explicit timezone - they are reproduced as written; absolute UTC alignment depends on the host timezone configuration at the time.

Timestamp	Source	Event	Confidence
2018-08-31 21:43:21	Amcache	c:\programdata\perfmon-k\perfmon-k.exe executed (SHA1 c31bbf518e9d9bc74a6567a2e3f0c37043ad399a)	Confirmed
2018-08-31 21:48:03	Amcache	c:\programdata\perfmon-k\perfmon-kconfigure.exe executed (SHA1 c95f567910b56d5e1f865781582aa6c8ff9f5db9)	Confirmed
2019-05-08 11:14:04 (host local)	McAfee OnAccessScanLog	McAfee deleted C:\ProgramData\perfmon-k\perfmon-kconfigure.exe - Artemis!1977C126E8E7 (MD5 1977c126e8e74da4460f41653ca6a6f2)	Confirmed
2019-05-17 18:29:13 (host local)	McAfee OnAccessScanLog	McAfee deleted C:\ProgramData\perfmon-k\perfmon-k.exe - Artemis!50CCE7D590CA (MD5 50cce7d590ca72cfb1f3b5522ea5c27b)	Confirmed

Timestamp	Source	Event	Confidence
2019-10-22 to 2019-12-19 (host local)	McAfee OnAccessScanLog	84 detections of MD5 264609c34194d47dbf25d18d4f83d41d (`{GUID}-78.0.3904.70_77.0.3865.120_chrome_updater.exe`) in mhill Local\Temp; classified GenericRXIT-PW (likely AV FP - see FND-007)	Confirmed event; Probable FP
2021-09-16 02:52:09.931	Sysmon EID 1	services.exe -> C:\Windows\subject_srv.exe -s "172.16.5.25:5682" -l 3262 -v "F-Response Subject" -k "155720749" (F-Response acquisition begins)	Confirmed
2021-09-16 02:52:10.873	Sysmon EID 11	subject_srv.exe creates C:\Windows\Mnemosyne.sys (F-Response kernel driver)	Confirmed
2021-09-16 03:01:58.401	Sysmon EID 1 / Prefetch	Second subject_srv.exe execution (run count = 2)	Confirmed
2021-09-16 03:10:21	Acquisition metadata	E01 image acquisition completed	Confirmed

Findings

FND-001 - Perfect Keylogger Installed at C:\ProgramData\perfmon-k

- **Confidence:** Confirmed
- **Claim type:** Direct Observation
- **Evidence source:** EVID-001
- **Tool output paths:** ./exports/perfmonk/install.log, ./analysis/parsed/install_bin_strings.txt, ./logs/20260604T201017Z_11740_stdout.log
- **Timestamp (UTC):** Directory mtime 2019-05-17 (post-AV cleanup); file content mtimes range 2017-08-31 to 2018-08-11.
- **Analyst interpretation:** The on-disk file install.log self-identifies the product as **BlazingTools Perfect Keylogger** with installation directory C:\ProgramData\perfmon-k. The directory name and file-name pattern (perfmon-k*) deliberately mimic Microsoft Performance Monitor (perfmon) - a defensive evasion tactic.
- **Limitation:** The specific Perfect Keylogger version was not extracted; this does not affect identification.

FND-002 - perfmon-k.exe and perfmon-kconfigure.exe Were Executed

- **Confidence:** Confirmed
- **Claim type:** Direct Observation
- **Evidence source:** EVID-001
- **Tool output paths:** ./analysis/parsed/amcache_UnassociatedFileEntries, ./logs/20260604T201423Z_10129_stdout.log
- **Timestamp (UTC):** 2018-08-31 21:43:21 and 2018-08-31 21:48:03
- **Analyst interpretation:** Amcache contains two records linking these executables (SHA1s c31bbf51... and c95f5679...) to C:\ProgramData\perfmon-k\ with associated File ID/Program ID metadata that on Windows 10 is consistent with documented execution-related recording semantics.
- **Limitation:** Sysmon log only covers from 2021-07-23, so no Sysmon corroboration is available for 2018 execution; relying on Amcache alone.

FND-003 - McAfee Removed Two Perfect Keylogger Components in 2019

- **Confidence:** Confirmed

- **Claim type:** Direct Observation
- **Evidence source:** EVID-001
- **Tool output paths:** ./analysis/parsed/mcafee_real_detections.txt
- **Timestamp (host local):** 2019-05-08 11:14:04 and 2019-05-17 18:29:13
- **Analyst interpretation:** McAfee VirusScan Enterprise (Engine 6100.8979) on-access scan detected and deleted both binaries as Artemis (cloud-heuristic) Trojan classifications. AV deletion confirms the binaries existed on disk and were considered malicious by the vendor cloud lookup at the time.
- **Limitation:** Artemis is a generic cloud-lookup classifier; the McAfee log itself does not establish the specific malware family. The corroborating identification as Perfect Keylogger comes from FND-001 (install.log).

FND-004 - Seven Perfect Keylogger Component Files Remain on Disk

- **Confidence:** Confirmed
- **Claim type:** Direct Observation
- **Evidence source:** EVID-001
- **Tool output paths:** ./logs/20260604T200936Z_31896_stdout.log, ./exports/perfmonk/
- **Timestamp (UTC):** Files present at 2021-09-16 (acquisition time); directory listing recovered from read-only mount.
- **Analyst interpretation:** Although McAfee removed perfmon-k.exe and perfmon-kconfigure.exe, the support files (recorder, viewer, hook, web, and data files) remained accessible at their original paths. These are residual artefacts of the keylogger installation.
- **Limitation:** McAfee on-access scan log captured ends 2021-07-22; AV state in the final ~2 months prior to acquisition is unknown.

FND-005 - No Execution of Residual Keylogger Files in Sysmon Window (Dormant)

- **Confidence:** Probable
- **Claim type:** Supported Inference
- **Evidence source:** EVID-001
- **Tool output paths:** ./analysis/parsed/sysmon.json (308,811 records), ./logs/20260604T201646Z_4111_stdout.log
- **Timestamp (UTC):** Sysmon window 2021-07-23 to 2021-09-16
- **Analyst interpretation:** A case-insensitive search for perfmon-kr|perfmon-kvw|perfmon-khk|perfmon-ki|perfmon-kwb across the entire Sysmon JSON returned zero hits. No registry persistence pointing at the perfmon-k directory was found in SAM, SOFTWARE, SECURITY, SYSTEM, DEFAULT, or any NTUSER.DAT (string sweep). Combined with the absence of these SHA-1 hashes in Amcache (only the long-deleted launchers were ever recorded), the residual files appear inert within the observable window.
- **Alternative explanation:** A keylogger may execute as injected code inside another process via DLL injection (e.g., perfmon-khk.dll loaded as a global hook), which would not necessarily produce a file-name hit in Sysmon event filters. This cannot be excluded without a deeper process-image/loaded-module review.
- **Limitation:** Inference from absence of evidence covers only 2021-07-23 to 2021-09-16; prior behavior is not assessed beyond Amcache.

FND-006 - subject_srv.exe Is F-Response (Legitimate DFIR Tool)

- **Confidence:** Confirmed
- **Claim type:** Direct Observation
- **Evidence source:** EVID-001
- **Tool output paths:** ./analysis/parsed/sysmon_subject_srv_first20.json, ./analysis/parsed/prefetch_parsed.csv
- **Timestamp (UTC):** 2021-09-16 02:52:09.931 and 03:01:58.401
- **Analyst interpretation:** Sysmon EID 1 records the command line C:\windows\subject_srv.exe -s "172.16.5.25:5682" -l 3262 -v "F-Response Subject" -k "155720749", parent process services.exe,

integrity level System. The same process subsequently created C:\Windows\Mnemosyne.sys (Sysmon EID 11) - the F-Response physical disk driver. Prefetch records two runs the same day. The host image was acquired 03:10 UTC the same day. This pattern matches F-Response Enterprise / Consultant Edition Subject Service behaviour for remote read-only forensic acquisition.

- **Limitation:** The F-Response engagement records (responder identity, authorization) are external to this evidence; the action is technically legitimate but the analyst cannot independently verify the engagement contractual basis.

FND-007 - Chrome Updater Detections Likely AV False Positives

- **Confidence:** Probable
- **Claim type:** Supported Inference
- **Evidence source:** EVID-001
- **Tool output paths:** ./analysis/parsed/mcafee_real_detections.txt, ./logs/20260604T201308Z_23781_stdout.log
- **Timestamp (host local):** 2019-10-22 to 2019-12-19 (84 events)
- **Analyst interpretation:** All 84 detections target the identical MD5 264609c34194d47dbf25d18d4f83d41d at the path C:\Users\mhill\AppData\Local\Temp\{GUID}-78.0.3904.70_77.0.3865.120_chrome_updater.exe. This file name pattern matches Google Chrome documented per-user auto-update installer behaviour. Amcache contains the corresponding legitimate Chrome 78.0.3904.70 installer in c:\program files (x86)\google\update\download\{8a69d345-d564-463c-aff1-a69d9e530f96}\78.0.3904.70\ (SHA1 30e8d3cf3a06da33c54b3d497bc6f16d50f578a4). The McAfee generic signature GenericRXIT-PW has historically produced false positives against legitimately signed Chrome installers.
- **Contradictory evidence:** The recorded parent process is svchost.exe, not GoogleUpdate.exe. This is consistent with a service-hosted task launching an installer, but does not by itself confirm legitimacy.
- **Alternative explanation:** Genuine bundled malware leveraging the Chrome update naming convention cannot be excluded without recovering and analysing the file body; McAfee deletion prevents this.
- **Limitation:** The deleted file is unavailable for independent analysis.

IOC Table

See ./analysis/ioc_register.csv for the full IOC register.

IOC ID	Type	Indicator	Confidence	Description
IOC-001	File path	C:\ProgramData\perfmon-k\	Confirmed	Perfect Keylogger install directory
IOC-002	SHA-256	fc7f5094bc182a05e782e0b775eccab9e662a3d207e89ab7f075db18a4c19c75	Confirmed	perfmon-kr.exe (recorder)
IOC-003	SHA-256	128cdb7c58bfda11a88c21e41e48df47f2122a1761d69aec5af79e241a712185	Confirmed	perfmon-kvw.exe (viewer)
IOC-004	SHA-256	225efff946151edeb6f860f5e4d68aa7025ca7abef67ddb215f5c89445d3356c	Confirmed	perfmon-khk.dll (keyboard hook)
IOC-005	SHA-256	fa76d9059fc4fe103e3de1868c76d86efbac9295fcd3e5f1c430cd1c5b4ad710	Confirmed	perfmon-ki.dll

IOC ID	Type	Indicator	Confidence	Description
IOC-006	SHA-256	8db5e3bdd57ed77f80e1dbd5f11a48a19f69faa2281f1039b615ee8b4a09407c	Confirmed	perfmon-kwb.dll
IOC-007	MD5	1977c126e8e74da4460f41653ca6a6f2	Confirmed	perfmon-kconfigure.exe (AV-deleted 2019-05-08)
IOC-008	MD5	50cce7d590ca72cfb1f3b5522ea5c27b	Confirmed	perfmon-k.exe (AV-deleted 2019-05-17)
IOC-013	MD5	264609c34194d47dbf25d18d4f83d41d	Probable false positive	Chrome 78 updater (see FND-007)
IOC-014	SHA-256	87c8fa606729ed63cb9d59f6b731338f8b06addbb3ef91e99b773eac2f2c524d	Confirmed legitimate	F-Response subject_srv.exe
IOC-015	Network	172.16.5.25:5682	Confirmed legitimate	F-Response examiner endpoint

On-Disk Component Table (Perfect Keylogger Residual Files)

File Path	Size Bytes	MD5	SHA-256	Evidence Source	Notes
C:\ProgramData\perfmon-k\perfmon-kr.exe	1352704	2d444c6afbfd647d246d2eb01a9e75c9	fc7f5094bc182a05e782e0b775eccab9e662a3d207e89ab7f075db18a4c19c75	EVID-001	PE32 GUI i386
C:\ProgramData\perfmon-k\perfmon-kvw.exe	1570816	2c96b03ff718bd2cdf4288864edeb6a0	128cdb7c58bfda11a88c21e41e48df47f2122a1761d69aec5af79e241a712185	EVID-001	PE32 GUI i386
C:\ProgramData\perfmon-k\perfmon-khk.dll	1307648	7cfcf2e7996ce96b44013e5fe0705820	225efff946151edeb6f860f5e4d68aa7025ca7abef67ddb215f5c89445d3356c	EVID-001	PE32 DLL GUI i386 (keyboard hook)
C:\ProgramData\perfmon-k\perfmon-ki.dll	275968	d567a6d0647f80ecb5a761ddd9ad367c	fa76d9059fc4fe103e3de1868c76d86efbac9295fcd3e5f1c430cd1c5b4ad710	EVID-001	PE32 console DLL i386
C:\ProgramData\perfmon-k\perfmon-kwb.dll	1348608	2ff0e7f97038863c4448bc08eb4286c8	8db5e3bdd57ed77f80e1dbd5f11a48a19f69faa2281f1039b615ee8b4a09407c	EVID-001	PE32 DLL GUI i386
C:\ProgramData\perfmon-k\install.bin	83456	a7e4adb5314851ff3899908875c0667c	SEE_APPENDIX	EVID-001	PE32 GUI i386 (installer)
C:\ProgramData\perfmon-k\install.log	520	116f70a01c6a10b2d36badc40d046e33	SEE_APPENDIX	EVID-001	Identifies BlazingTools Perfect Keylogger
C:\ProgramData\perfmon-k\opt.dat	64	7f73f5c6669f455d39564da49c2199b8	SEE_APPENDIX	EVID-001	Data file
C:\ProgramData\perfmon-k\pkl.bin	9368	7df7f4e9098ea82467e01d4702b678fa	SEE_APPENDIX	EVID-001	Data file

File Path	Size Bytes	MD5	SHA-256	Evidence Source	Notes
C:\ProgramData\perfmon-k\web.dt	304	d22de33006cf9b95f2de4db0d587b5cd	SEE_APPENDIX	EVID-001	Data file
C:\ProgramData\perfmon-k\help.chm	407279	a3838f8bbf585d3c5cfc3b615fcc50c9	SEE_APPENDIX	EVID-001	Product help file (contains Perfect Keylogger string)

AV-Deleted Components (Not Recovered)

File Path	Reported MD5	Detection	Action Time (host local)	Evidence Source
C:\ProgramData\perfmon-k\perfmon-kconfigure.exe	1977c126e8e74da4460f41653ca6a6f2	Artemis!1977C126E8E7 (Trojan)	2019-05-08 11:14:04 DELETED_NOT_RECOVERED	McAfee OnAccessScanLog
C:\ProgramData\perfmon-k\perfmon-k.exe	50cce7d590ca72cfb1f3b5522ea5c27b	Artemis!50CCE7D590CA (Trojan)	2019-05-17 18:29:13 DELETED_NOT_RECOVERED	McAfee OnAccessScanLog

Claim Support Matrix

Finding ID	Claim	Claim Type	Confidence	Supporting Evidence	Tool Output Path	Alternative Explanation	Limitation
FND-001	Perfect Keylogger installed at C:\ProgramData\perfmon-k	Direct Observation	Confirmed	install.log string + component file layout	./exports/perfmonk/install.log	Another product reusing boilerplate (rejected by file-name pattern)	Specific PK version not extracted
FND-002	perfmon-k.exe and perfmon-kconfigure.exe were executed 2018-08-31	Direct Observation	Confirmed	Amcache records with FileID/ProgramID	./analysis/parsed/amcache_UnassociatedFileEntries	Amcache inventory-only entries (low likelihood on Win10)	No 2018 Sysmon coverage
FND-003	McAfee deleted 2 PK components in 2019 (Artemis Trojan)	Direct Observation	Confirmed	McAfee OnAccessScanLog	./analysis/parsed/mcafee_real_detections.txt	Cloud-heuristic FP (mitigated by FND-001 corroboration)	Binaries not recovered
FND-004	7 PK component files remain on disk at acquisition	Direct Observation	Confirmed	Filesystem listing	./logs/20260604T200936Z_31896_stdout.log	Quarantine container (rejected - original paths)	McAfee log ends 2021-07-22
FND-005	No PK execution in Sysmon window (dormant)	Supported Inference	Probable	Zero Sysmon hits across 308,811 events; no registry persistence found	./analysis/parsed/sysmon.json	Injected DLL hook would not name-match	Pre-2021-07-23 not covered

Finding ID	Claim	Claim Type	Confidence	Supporting Evidence	Tool Output Path	Alternative Explanation	Limitation
FND-006	subject_srv.exe is F-Response (legitimate DFIR tool)	Direct Observation	Confirmed	Sysmon command line shows -v F-Response Subject; Mnemosyne.sys creation	./analysis/parsed/sysmon_subject_srv_first20.json	Threat-actor mimicry (rejected - command-line and behavior match documented F-Response)	Engagement records external to evidence
FND-007	Chrome 78 updater detections likely McAfee false positives	Supported Inference	Probable	Path pattern matches Chrome auto-update; Amcache shows legit Chrome 78 installer	./analysis/parsed/mcafee_real_detections.txt	Genuine bundled malware (cannot fully exclude)	Deleted file unavailable for analysis

Persistence and Configuration Sweep (Results)

Check	Method	Result
HKLM\Software\Microsoft\Windows\CurrentVersion\Run	strings on SOFTWARE hive	No perfmon-k match
HKCU per-user Run keys (all NTUSER.DAT hives)	strings sweep	No perfmon-k match
HKLM\System services	strings on SYSTEM hive	No perfmon-k match
Scheduled Tasks (C:\Windows\System32\Tasks, C:\Windows\Tasks)	grep -r	No perfmon-k match
Default user hive	strings on DEFAULT	No perfmon-k match

Note: This sweep used string searches and is sensitive to non-obfuscated values; it does not exclude unconventional persistence such as WMI permanent subscriptions, COM object hijacking, or DLL search-order hijack. See Gaps and Limitations.

Gaps and Limitations

- **Sysmon coverage:** Only 2021-07-23 to 2021-09-16 covered. No telemetry for the historical install (2018) or the AV-removal events (2019).
- **Amcache:** Records execution-related metadata but does not guarantee a process was launched in every case; semantics vary by Windows build.
- **McAfee log timestamps:** Recorded in apparent host local time without explicit timezone. UTC alignment relies on the host timezone setting at the time of logging, which was not extracted.
- **AV state late 2021:** McAfee on-access scan log ends 2021-07-22; whether the AV remained fully active through 2021-09-16 was not independently confirmed.
- **Persistence search:** String-based sweep across the standard hives and the Tasks directories. WMI permanent event subscriptions, COM hijacks, DLL search-order persistence, AppInit_DLLs (encoded), and Image File Execution Options were not parsed by dedicated tools.
- **Deleted binaries:** The two AV-removed binaries are not present on disk and were not recovered for independent analysis.
- **Memory:** No memory image was provided; in-process keylogger activity (e.g., loaded perfmon-khk.dll via global hook) cannot be assessed.
- **MFT/USN journal:** Not parsed in this report; would corroborate file-system creation/deletion events.

- **PowerShell, Security, WMI Activity event logs:** Captured but not exhaustively analysed for unrelated incidents.

Unknowns

- **What is known:** A commercial keylogger was installed in 2018, partially removed by AV in 2019, and its residual components remained on disk at acquisition; no in-window keylogger execution observed.
- **What is unknown:** Initial install vector; principal user impacted by the keylogger collection (no recovered keylog file); whether any captured data was exfiltrated before AV removal; current execution state via injection.
- **What cannot be determined from available evidence:** Whether the deleted binaries had any data-exfiltration capability that actually completed against this host.
- **Evidence that would resolve uncertainty:** Memory image of the host; off-host network telemetry from 2018-2019; recovery of the deleted launcher binaries; review of MFT/USN for keylogger output files; WMI subscription parser output; review of NTUSER hives via a dedicated registry parser.

Reproducibility Appendix

- **Working directory:** /home/sansforensics/Desktop/cases/Test1
- **Command ledger:** ./logs/command_ledger.csv
- **Total commands logged:** 99
- **Failed commands:** 22 (all are diagnostic dead-ends: e.g., mmls on a partition-table-less volume image, missing PECmd/yara installs, SIGPIPE from head, and absence-confirming string searches that return non-zero when no match. None are forensic data losses.)
- **Rerun or retry commands noted:** 1
- **Unresolved command failures:** 0
- **Command ledger audit:** PASSED (audit_command_ledger.py output: Command ledger audit passed.)
- **Claim validation:** PASSED (validate_claims.py output: Claim validation passed. Findings checked: 7)
- **Artifact manifest:** ./reports/artifact_manifest.csv (704 file entries with SHA-256 hashes)

Reproducibility - Mount & Verification Commands

```
ewfinfo /home/sansforensics/Desktop/cases/Test1/base-wkstn-01-c-drive.E01
ewfverify /home/sansforensics/Desktop/cases/Test1/base-wkstn-01-c-drive.E01
sudo ewfmount /home/sansforensics/Desktop/cases/Test1/base-wkstn-01-c-drive.E01 /mnt/ewf_wkstn01
sudo mount -o ro,loop,noatime /mnt/ewf_wkstn01/ewf1 /mnt/wkstn01
sudo fsstat /mnt/ewf_wkstn01/ewf1
```

Reproducibility - Artifact Extraction (selected)

```
sudo cp /mnt/wkstn01/Windows/Prefetch/*.pf ./exports/prefetch/
sudo cp /mnt/wkstn01/Windows/System32/winevt/Logs/*.evtx ./exports/evtx/
sudo cp /mnt/wkstn01/Windows/System32/config/SOFTWARE ./exports/registry/
sudo cp /mnt/wkstn01/Windows/appcompat/Programs/Amcache.hve ./exports/amcache/
sudo cp /mnt/wkstn01/Users/mhill/NTUSER.DAT ./exports/registry/mhill_NTUSER.DAT
sudo cp /mnt/wkstn01/ProgramData/perfmon-k/* ./exports/perfmonk/
sudo cp /mnt/wkstn01/Windows/subject_srv.exe ./exports/binaries/
sudo cp /mnt/wkstn01/ProgramData/McAfee/DesktopProtection/*.txt ./exports/mcafee/
sudo cp /mnt/wkstn01/ProgramData/McAfee/Agent/logs/*.log ./exports/mcafee/
```

Reproducibility - Parsing Commands

```
python3 ./analysis/parse_prefetch.py ./exports/prefetch ./analysis/parsed/prefetch_parsed.csv
dotnet /opt/zimmermantools/EvtxeCmd/EvtxECmd.dll -f ./exports/evtx/Microsoft-Windows-
Sysmon%4Operational.evtx --json ./analysis/parsed --jsonf sysmon.json
dotnet /opt/zimmermantools/AmcacheParser.dll -f ./exports/amcache/Amcache.hve --csv ./analysis/
parsed --csvf amcache -i
dotnet /opt/zimmermantools/EvtxeCmd/EvtxECmd.dll -f ./exports/evtx/Microsoft-Windows-Windows
Defender%4Operational.evtx --csv ./analysis/parsed --csvf defender.csv
dotnet /opt/zimmermantools/EvtxeCmd/EvtxECmd.dll -f ./exports/evtx/Security.evtx --csv ./analysis/
parsed --csvf security.csv
```

Confidence Level Definitions

Confidence	Definition
Confirmed	Directly supported by raw or parsed forensic artifact.
Corroborated	Supported by two or more independent artifacts.
Probable	Supported by one artifact plus strong contextual evidence.
Hypothesis	Plausible but not yet proven; must include the next validation step.
Rejected	Tested and unsupported.